DAY 06 – IPV4

IPv4 THEORY (10 marks)

1. What is IPv4?
    - IPv4 is an Internet Protocol version 4.
    - Provides Logical Addressing for device communicating over IP networks.

2. How many bits are in an IPv4 address: 32 bits.

3. What is an octet?
    - An Octet divides 32-bit address into 8-bit section.

4. How many octets are in an IPv4 address: 8 bits

5. What is a subnet mask?
    - subnet mask is a 32-bit number that splits an IPv4 address into a network ID and a host ID.

6. Difference between network portion and host portion?

    Network Portion:
    - Identifies the group or network.
    - Same for all devices on same network

    Host Portion:
    - Identifies a specific device inside that network
    - Unique to each Individual device

7. What is a private IPv4 address?
    - The main purpose of private IPv4 addresses is to allow devices to communicate within a private network while saving limited public IP addresses.

8. Write all three private IPv4 ranges.
    - Class A, B, C

9. What is a public IPv4 address?
    - Class D, E

10. What is the IPv4 loopback address?
    - It is mainly used for checking Network stack with the range of 127.0.0.0
    - a reserved IP address that lets a device send and receive data to and from itself without passing through an external network or physical cable.

DAY 06 – IPV4

11. What is APIPA?
    - APIPA stands for Automatic private Ip addressing
    - It automatically assigns itself an IPV4 address when it cannot reach a DHCP server.

12. What does 169.254.x.x commonly indicate?
    - A 169.254.x.x address indicates that a device could not obtain an IP address from a DHCP server and automatically assigned itself an APIPA address.

13. What is a default gateway?
    - Router sends traffic outside the Local Network.

14. Static vs dynamic IP addressing?

    Static IP addressing:
    - Manually Configured
    - Kept consistent
    - Used in servers, printers et…

    Dynamic IP addressing:
    - Assigned automatically by DHCP
    - Provides IPV4 addresses, subnet mask, gateway and DNS.

15. What is DHCP?
    - DHCP stands for Dynamic Host configuration Protocol
    - It is a network protocol that assigns IP addresses and network settings to devices when they connect to a network.

16. What is a network address?
    - A network address is a unique identifier used to locate a specific node, host, or entire subnetwork on a computer or telecommunications network.

17. What is a broadcast address?
    - A broadcast address is a special networking address used to send data simultaneously to all devices connected to a local network segment or subnet.

18. Why is IPv4 knowledge important in cybersecurity?
    - IPv4 knowledge is fundamental to cybersecurity because it serves as the backbone of modern network communication and security architecture.

DAY 06 – IPV4

PRACTICAL IDENTIFICATION (4 marks)

Your own: IPv4 address, Subnet mask, Default gateway, DNS server, DHCP status.

IPV4 Address: 10.101.139.72

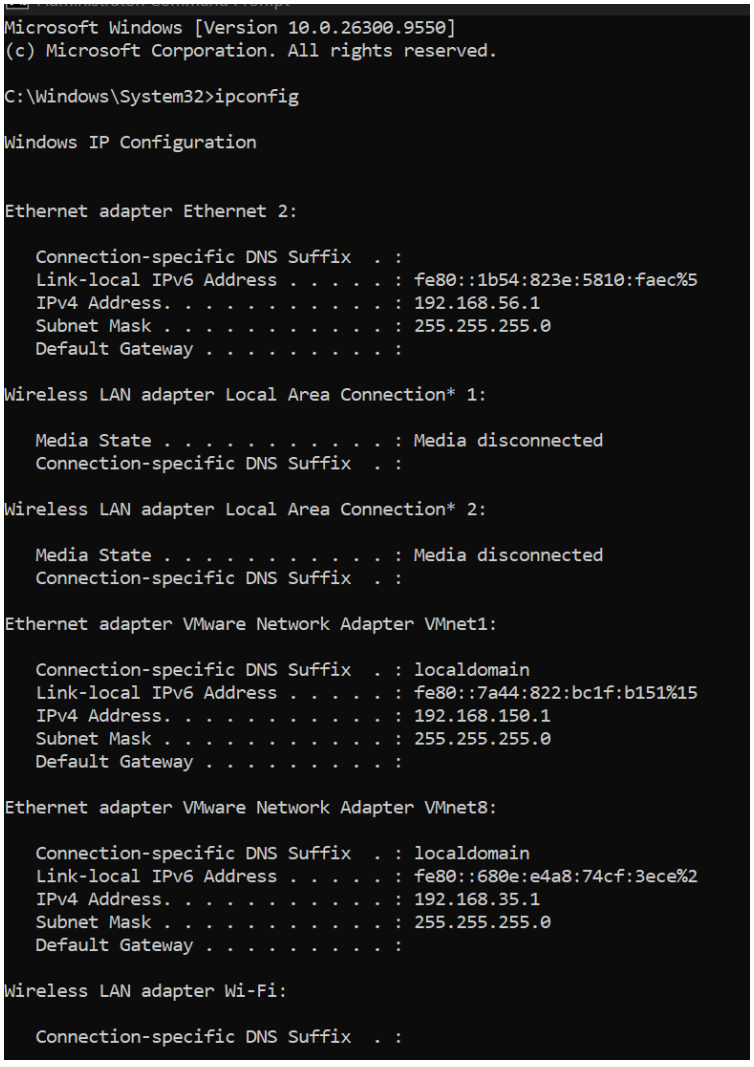
Subnet Mask: 255.255.224.0

Default Gateway: 10.101.128.1

DHCP status: 9.9.9.9, 208.67.222.222, 8.8.8.8.

SCREENSHOTS (3 marks)

Attach screenshots of: ipconfig, ipconfig /all, ping 127.0.0.1



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Microsoft Windows [Version 10.0.26300.9550]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1b54:823e:5810:faec%5
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . : localdomain
   Link-local IPv6 Address . . . . . : fe80::7a44:822:bc1f:b151%15
   IPv4 Address. . . . . . . . . . . : 192.168.150.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . : localdomain
   Link-local IPv6 Address . . . . . : fe80::680e:e4a8:74cf:3ece%2
   IPv4 Address. . . . . . . . . . . : 192.168.35.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
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DAY 06 – IPV4

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   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::450d:aba6:dee6:4cd7%18
   IPv4 Address. . . . . . . . . . . : 10.101.139.72
   Subnet Mask . . . . . . . . . . . : 255.255.224.0
   Default Gateway . . . . . . . . . : 10.101.128.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Windows\System32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Kirthikaa
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-05
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::1b54:823e:5810:faec%5(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 621412391
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
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DAY 06 – IPV4

```
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 26-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 2A-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . : localdomain
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::7a44:822:bc1f:b151%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.150.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 12 October 2026 9.08.30 AM
   Lease Expires . . . . . . . . . . : 12 October 2026 11.54.37 AM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.150.254
   DHCPv6 IAID . . . . . . . . . . . : 201347158
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   DNS Servers . . . . . . . . . . . : 192.168.150.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . : localdomain
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet8
   Physical Address. . . . . . . . . : 00-50-56-C0-00-08
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::680e:e4a8:74cf:3ece%2(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.35.1(Preferred)
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```
   IPv4 Address. . . . . . . . . . . : 192.168.35.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 12 October 2026 9.08.30 AM
   Lease Expires . . . . . . . . . . : 12 October 2026 11.54.37 AM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.35.254
   DHCPv6 IAID . . . . . . . . . . . : 234901590
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   DNS Servers . . . . . . . . . . . : 192.168.35.2
   Primary WINS Server . . . . . . . : 192.168.35.2
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::450d:aba6:dee6:4cd7%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.101.139.72(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.224.0
   Lease Obtained. . . . . . . . . . : 12 October 2026 9.08.44 AM
   Lease Expires . . . . . . . . . . : 12 October 2026 3.08.47 PM
   Default Gateway . . . . . . . . . : 10.101.128.1
   DHCP Server . . . . . . . . . . . : 10.101.128.1
   DHCPv6 IAID . . . . . . . . . . . : 153399993
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   DNS Servers . . . . . . . . . . . : 9.9.9.9
                                       208.67.222.222
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-76
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Windows\System32>ping 127.0.0.1
```

DAY 06 – IPV4

```
C:\Windows\System32>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Windows\System32>_
```